

**NORTHERN UNIVERSITY OF BUSINESS
AND TECHNOLOGY, KHULNA**

WEEK 09

PROGRESS AND INTEGRATION REPORT

AI Smart Campus System

CSE 4204 Mobile Computing Lab | Section 8A | Team 07

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1. Project Overview

The AI Smart Campus System is an academic management and student-success platform for Northern University of Business and Technology, Khulna. The current implementation integrates a React/Vite frontend, Laravel REST API, MySQL database, role-based authentication, academic monitoring, campus services, and server-side AI workflows. Week 09 focuses on moving from separate modules to an integrated, demonstrable software system.

CURRENT STAGE	Major features integrated; formal testing and refinement remain
LIVE DEPLOYMENT	Available on Render
SOURCE CONTROL	Public GitHub repository with continuous development activity

Live Demo: <https://ai-smart-campus-system-ce9i.onrender.com/>

GitHub Repository: <https://github.com/nazmusshakib878/CSE4204-8A-T07-ai-smart-campus-system>

2. Summary of Completed Features

- Authentication workflow: registration, approval-aware login, password reset, logout, protected access, and profile management.
- Student workspace: dashboard, enrolled courses, attendance, grades, performance, class schedule, tasks, resources, notices, messages, campus services, support tickets, library/fee views, and course recommendations.
- Faculty workspace: dashboard, assigned-course workspace, student enrollment, attendance, assessment, grade/performance entry, student monitoring, risk indicators, notices, and permitted campus operations.
- Administrator workspace: user approval/account management, departments, courses, notices, routines, exams, events, fees, academic management, and institution-wide monitoring.
- AI-assisted workflows: conversational assistant, AI-assisted course recommendation ranking with deterministic fallback, and academic risk analysis with structured output.
- Database-backed academic and campus workflows integrated through the backend API rather than relying only on static UI data.

3. Partially Completed / Remaining Work

No major core module is currently marked Not Started. Remaining items are verification, negative-path evidence, repository cleanup, and formal testing/refinement activities.

- Cross-role authorization and protected-route verification are documented; only final negative-path database/network evidence remains.
- Mobile responsive evidence has been captured from the live deployment; additional tablet/breakpoint checks remain refinement work.
- Continue API failure, database failure, network interruption, quota/rate-limit, timeout, and invalid-input testing.
- Perform final UI consistency pass for loading, success, error, and empty states.

- Final report documentation and screenshot review are complete; only submission packaging and repository-side final checks remain.

4. Frontend-Backend-Database Integration Status

Major frontend modules communicate with the Laravel REST API for authentication, user/role data, courses, academic records, monitoring, campus services, notices, messaging, recommendations, and AI operations. Laravel Sanctum is used for protected API routes. MySQL is the default database, with migrations, seeders, models, and database-backed context used by the application and AI services.

Integration Area	Status	Week 09 Verification Focus
Authentication & sessions	Completed	Registration, login, logout, password reset, protected routes
Academic data	Completed	Courses, attendance, assessment, grades, CGPA/performance
Campus services	Completed	Routines, exams, events, fees, tickets, library, notices
Role-based workflows	Completed	Student, faculty and administrator workspaces
AI endpoints	Completed	Assistant, recommendations, academic risk analysis
Failure/edge-case coverage	In Progress	Provider errors, network/database failures, authorization edge cases
Mobile validation	Completed	Live mobile responsive evidence included; additional tablet checks are refinement.

5. AI Integration Status

The project contains three backend-controlled AI-assisted workflows. AI credentials, trusted context, authorization, retry logic, and response validation remain on the Laravel backend instead of being exposed to the browser.

- Student AI Assistant using Google Gemini for academic/campus guidance.
- Gemini-assisted course recommendation ranking restricted to known eligible course candidates, with deterministic fallback when the provider is unavailable.
- OpenAI-assisted academic risk analysis for authorized faculty/administrators using structured output and database-backed baseline indicators.
- Safe user-facing failure behavior for provider timeouts, quota/rate-limit issues, invalid responses, or unavailable configuration.

User -> React Frontend -> Laravel API -> Authentication/Validation -> Database Context -> AI Provider -> Validated Result -> UI

6. Authentication and Access Control

- Registration and approval-aware login flow.
- Password reset and password handling through backend-controlled authentication.
- Laravel Sanctum protection for authenticated API routes.
- Role-specific navigation and access for student, faculty, and administrator users.
- Protected-route and role-authorization verification is documented; invalid-session and negative-path checks remain part of final testing.

7. Major End-to-End User Workflows

7.1 Student Workflow

Register -> Approval/Login -> Student Dashboard -> Courses/Attendance/Grades -> Campus Services -> Course Recommendations / AI Assistance -> Result or Action Display

7.2 Faculty Workflow

Login -> Faculty Dashboard -> Assigned Courses -> Student Monitoring -> Attendance/Assessment/Grades -> Risk Analysis -> Notices/Campus Operations -> Result Display

7.3 Administrator Workflow

Login -> Admin Workspace -> User Approval/Management -> Department/Course/Academic Management -> Routines/Fees/Services -> Risk Monitoring -> Notices/System Data

8. UI/UX Improvements

- Consistent sidebar navigation and role-specific workspaces.
- Card-based dashboards, tables, filters, charts, forms, and status indicators.
- Validation messages on authentication forms and clear empty states in data views.
- Loading/thinking states for AI interactions and structured result presentation.
- Unified visual language across academic management, monitoring, messaging, services, recommendations, and profile pages.
- Live phone-responsive evidence is included in the screenshot section; tablet and additional workspace checks are treated as refinement/testing.

9. Error Handling and Reliability

- Input validation is performed before protected backend operations.
- Unauthorized requests are rejected by authentication/role checks.
- AI provider failures return safe user-facing errors rather than exposing credentials or raw provider details.
- Gemini course recommendation has a deterministic fallback to prevent an empty recommendation experience when the provider fails.
- AI retry handling covers temporary quota/rate-limit, timeout, and selected server-side failures.
- Formal Week 09 testing still needs explicit evidence for database unavailability and network interruption scenarios.

10. Major Problems Encountered and Solutions

Problem / Risk	Implemented Approach
Keeping frontend data synchronized with real backend state	Moved major workflows to REST API and database-backed data instead of isolated dummy UI data.
Different permissions for students, faculty and administrators	Implemented role-aware navigation, protected routes and scoped backend authorization.
AI service quota, timeout and temporary provider failures	Added retry/error handling, safe responses and deterministic fallback where appropriate.
Preventing AI from inventing or modifying authoritative academic records	AI uses trusted backend context; generated output is advisory and validated before display/persistence.
Managing many academic and campus modules consistently	Used shared dashboard, table, form, card and status patterns across role workspaces.

11. Review of Project Scope and Changes from Original Plan

The implementation remains centered on the approved AI Smart Campus System scope: a unified, role-based academic management and student-support platform integrating React/Vite, Laravel, MySQL, Google Gemini, and OpenAI. The major functional scope has not been reduced. The documented changes below clarify implementation choices that differ from the original proposal wording.

11.1 Changes from Original Plan

Original Plan	Current Implementation	Why It Changed	New Approach
JWT-based authentication	Laravel Sanctum Bearer-token authentication	Sanctum is Laravel-native and integrates cleanly with protected REST API routes.	Authenticated requests use Laravel Sanctum personal access tokens with role/approval checks.
Single AI assistant concept centered on Gemini	Gemini is used for the Student AI Assistant/course recommendation; OpenAI is used for structured academic-risk analysis.	Different AI tasks benefit from different provider capabilities and output formats.	AI providers remain server-side, use trusted database context, and return validated/advisory results.
Broad campus-information chatbot wording	Student AI Assistant is constrained to available trusted academic/campus context and does not invent unavailable university facts.	This reduces hallucination risk and protects authoritative campus information.	Unavailable information is handled safely instead of generating unsupported university-specific answers.

12. Testing and Verification Status

Automated verification is documented for the backend and AI workflows. The report records a focused AI suite with 15 passing tests and 71 assertions, and a full backend suite with 78 passing tests and 381 assertions. Frontend verification commands are also documented for unit tests, linting, production build, and end-to-end testing. Explicit database-unavailable/network-interruption evidence remains the only negative-path evidence item still marked In Progress in the feature checklist.

- New user registration and approval-aware login
- Existing user login and logout
- Protected routes and role authorization
- Academic and campus service operations
- AI assistant request/response and error paths
- Course recommendation provider fallback
- Academic risk analysis structure and persistence
- Invalid form submissions and safe error states

- Responsive interface and navigation (live mobile evidence captured; additional breakpoint checks continue as refinement)

13. Feature Completion Checklist

Major Feature	Status	Reason if Incomplete	Planned Completion
Public landing page	Completed		-
Registration	Completed		-
Login / Logout	Completed		-
Password reset	Completed		-
Role-based protected access	Completed		-
Student dashboard	Completed		-
Faculty dashboard	Completed		-
Administrator workspace	Completed		-
User management / approval	Completed		-
Academic data management	Completed		-
Attendance & performance monitoring	Completed		-
Course management / enrollment	Completed		-
Assessment / grades / CGPA	Completed		-
Class & exam routines	Completed		-
Campus services	Completed		-
Fees / library / support tickets	Completed		-
Notices / messages	Completed		-
Course recommendations	Completed		-
AI Assistant	Completed		-
Academic risk analysis	Completed		-
AI provider failure handling	Completed		-

Major Feature	Status	Reason if Incomplete	Planned Completion
API / database failure scenario evidence	In Progress	AI/API failure handling is documented; explicit database-unavailable/network-interruption evidence remains.	Before Week 09 review / testing phase
Cross-role authorization edge-case test	Completed	Protected-route and role-authorization verification is documented in the backend testing status.	-
Mobile responsiveness evidence	Completed	Live mobile responsive screenshots included in this report.	-
Final Week 09 report documentation cleanup	Completed	Report content, figure captions, responsive evidence and backend evidence were finalized.	-

14. GitHub Repository and Deployment Status

The project repository is public and organized into implementation and documentation areas. At the time of Week 09 report preparation, the repository shows continuous recent development activity. Major folders include backend, frontend, database, documentation, screenshots, and scripts.

Item	Current Evidence
GitHub Repository	Public repository available
Development Activity	Continuous recent commits visible in the repository/Insights activity
Backend	Laravel REST API, services, tests, migrations, seeders
Frontend	React/Vite application and frontend tests
Database	MySQL configuration plus database folder/backup notes
AI	Gemini assistant/recommendation + OpenAI risk analysis
Documentation	API, deployment, ER diagram, AI guide, Postman and screenshot checklist
Live Demo	Render deployment URL available

Repository: <https://github.com/nazmusshakib878/CSE4204-8A-T07-ai-smart-campus-system>

Live Demo: <https://ai-smart-campus-system-ce9i.onrender.com/>

15. Current Project Screenshots

The screenshots below document the current implementation. The newly supplied GitHub, database, and mobile screenshots are placed exactly as provided, without cropping, retouching, enhancement, or content replacement.

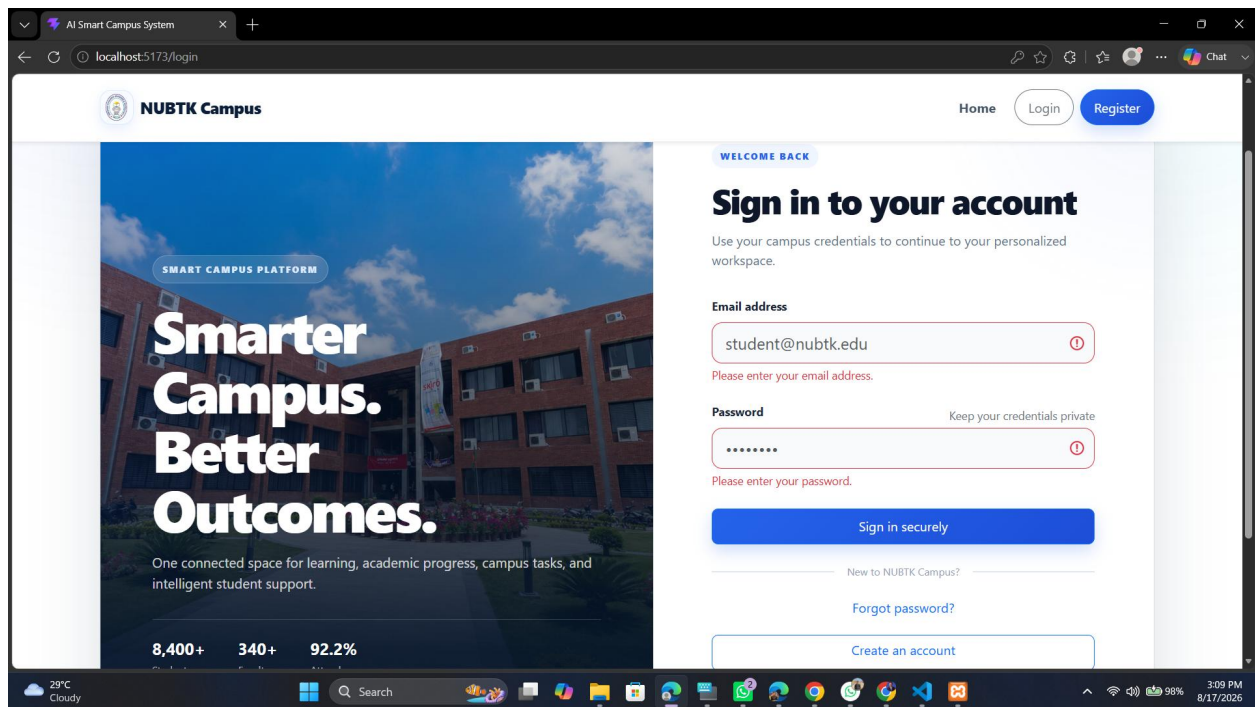


Figure 1. Login interface with client-side validation and secure sign-in workflow.

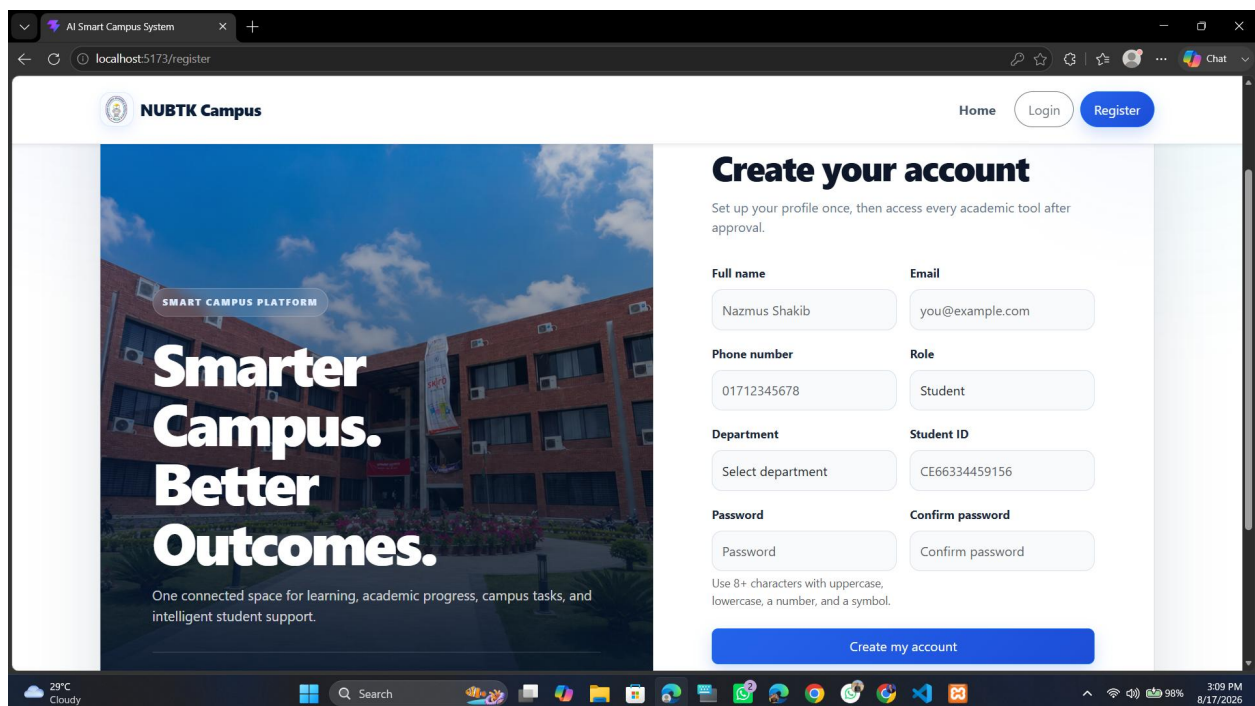


Figure 2. User registration interface.

15.1 Dashboard and Academic Workspaces

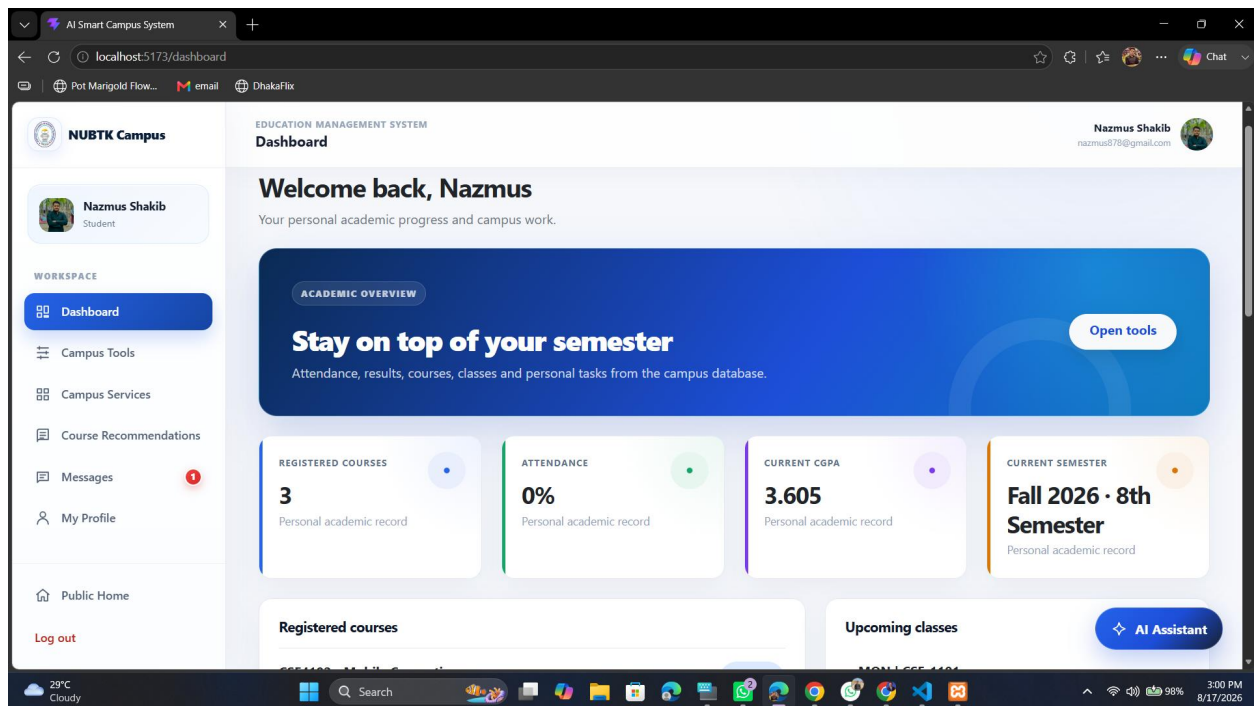


Figure 3. Student dashboard showing integrated academic and campus information.

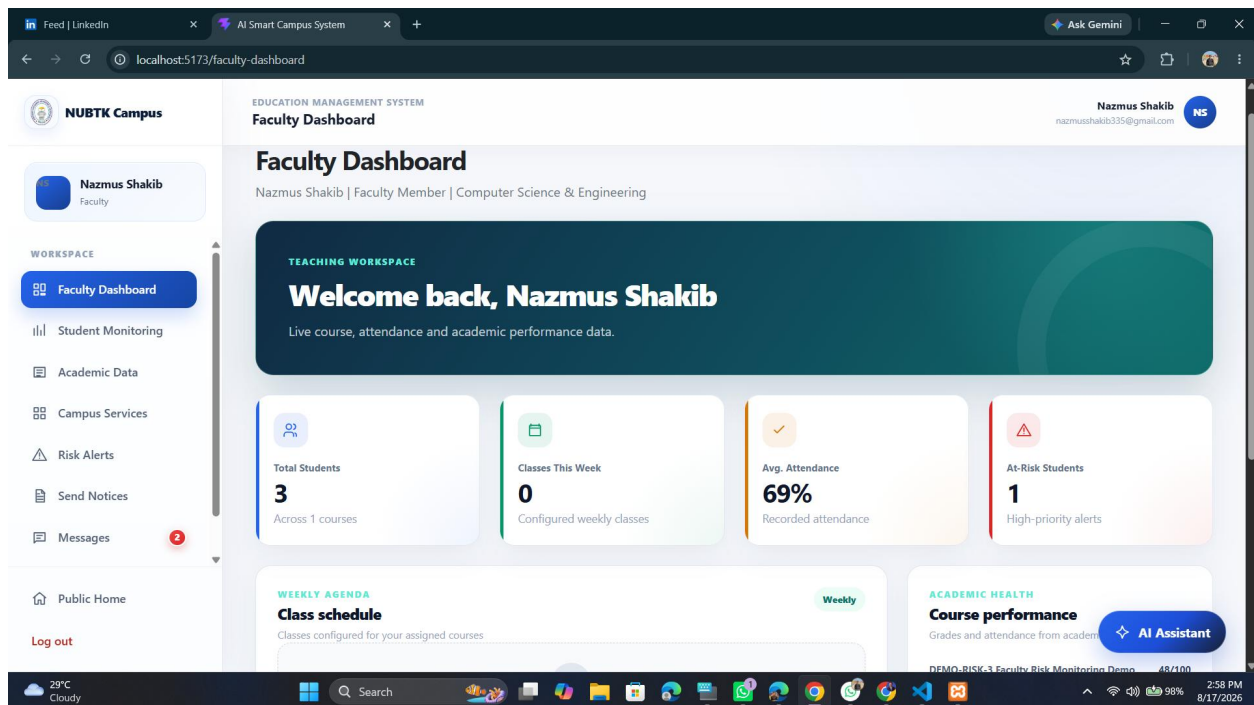


Figure 4. Faculty dashboard and academic monitoring workspace.

15.2 Administrative and Academic Management

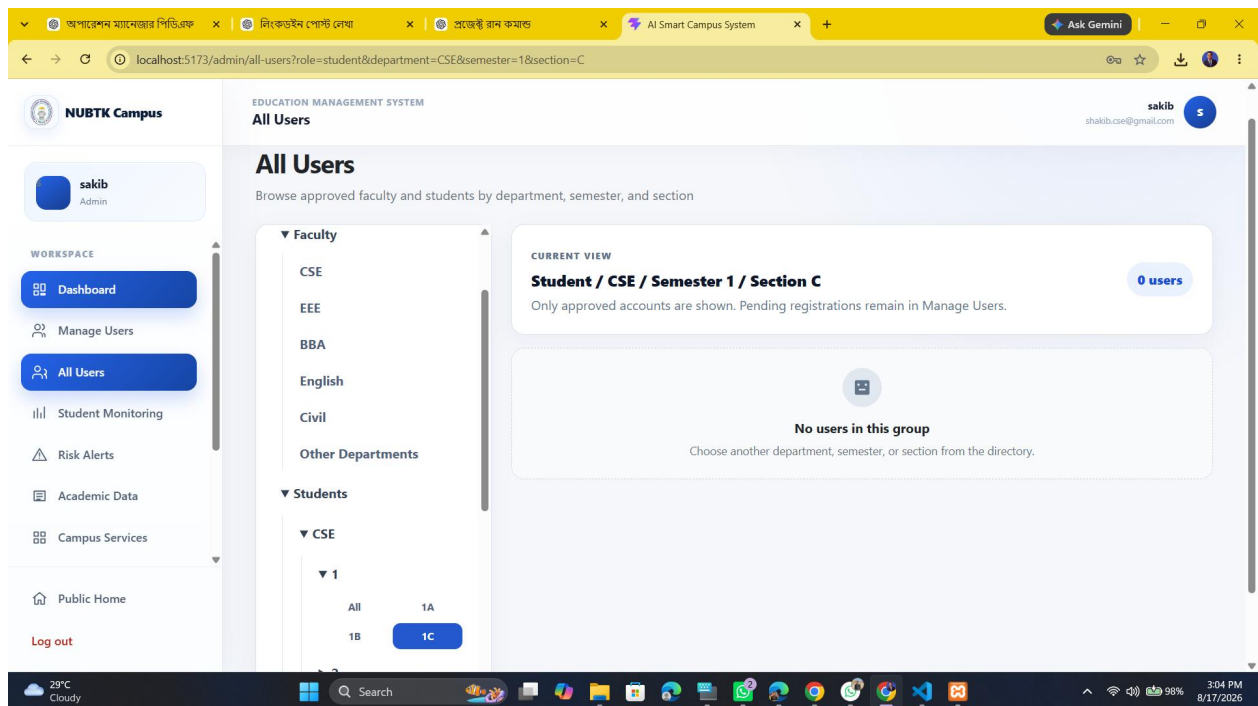


Figure 5. Administrator user directory/filtering view with empty-state handling.

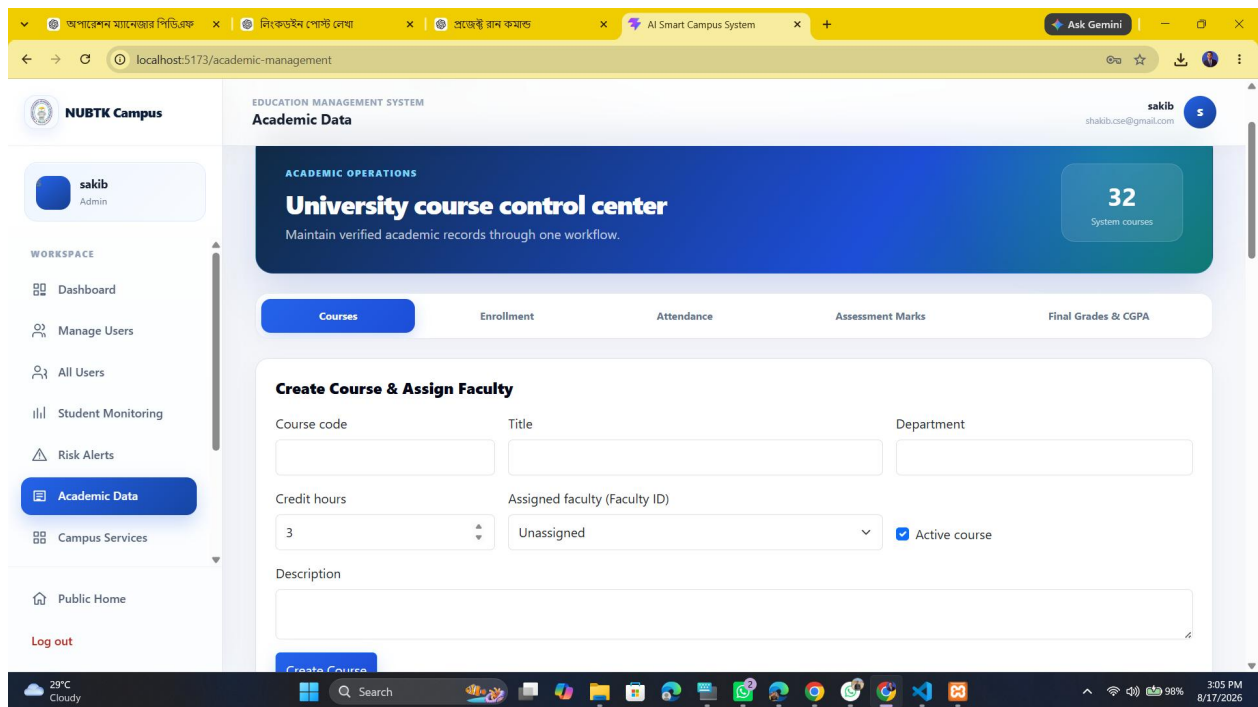


Figure 6. Academic management workspace backed by system data.

15.3 Recommendations and Campus Services

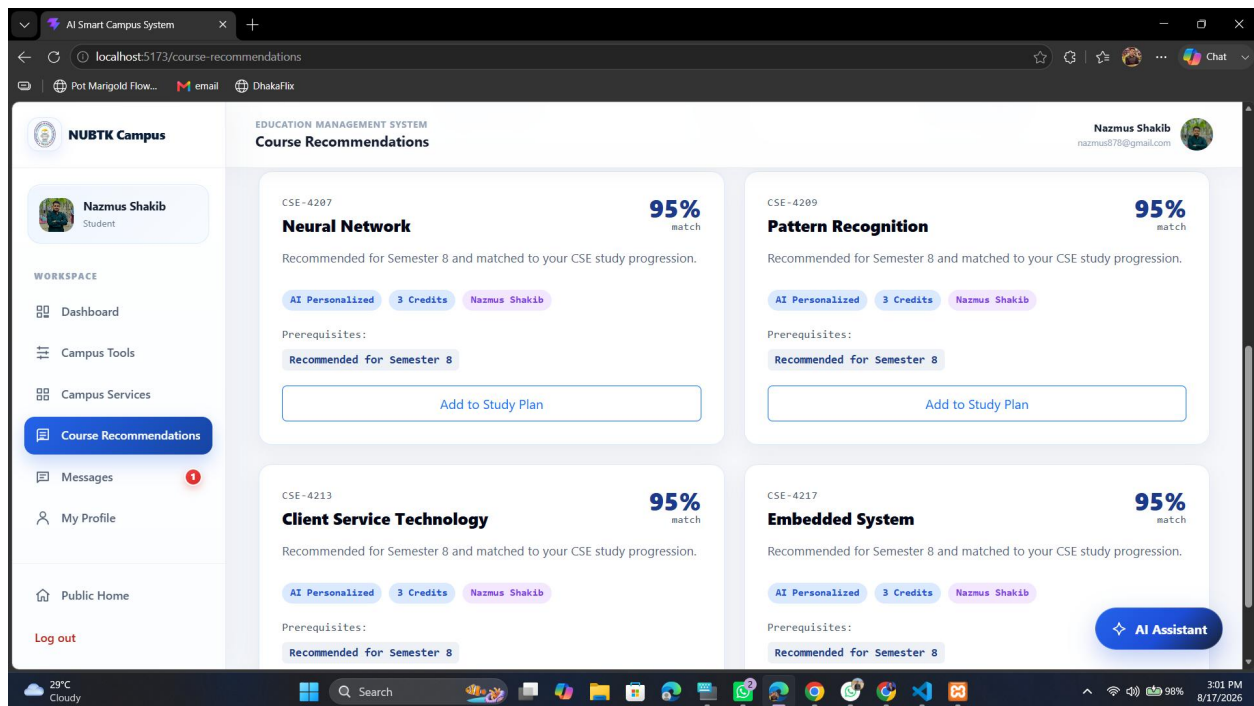


Figure 7. Course recommendation module integrated into the student workflow.

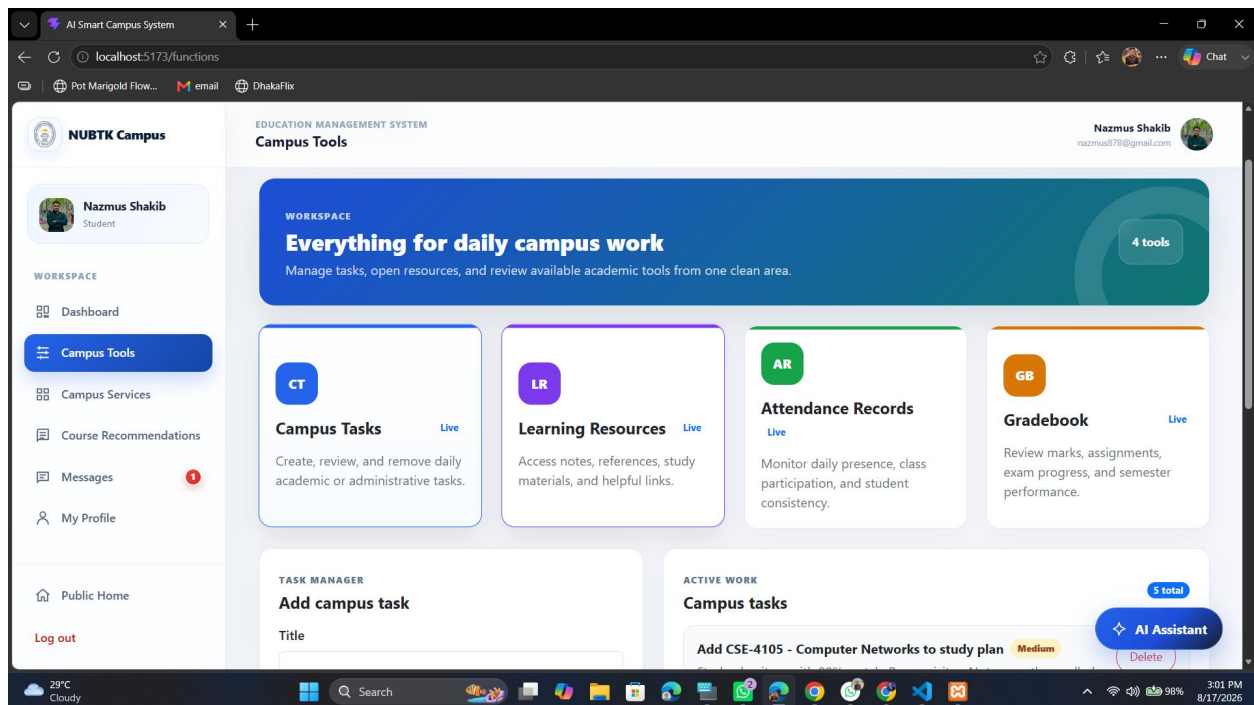


Figure 8. Campus services workspace and connected service modules.

15.4 AI and Risk Analysis

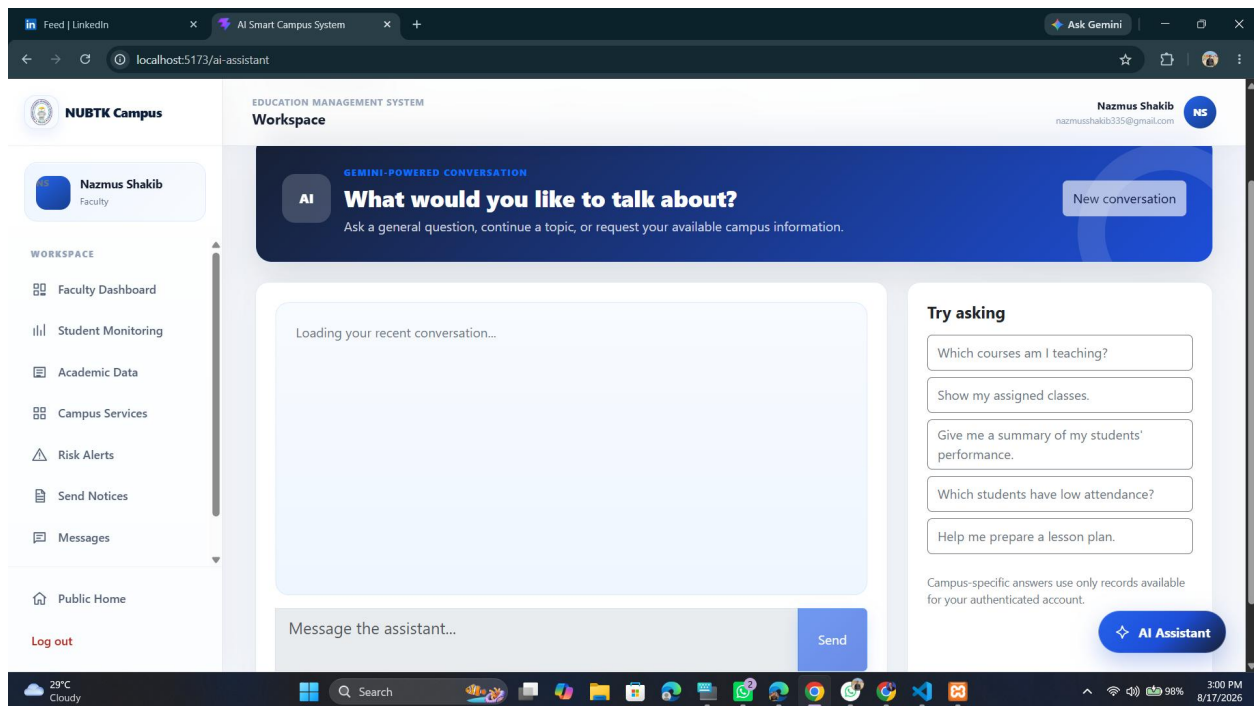


Figure 9. Student AI Assistant interface showing the application-to-AI interaction workflow.

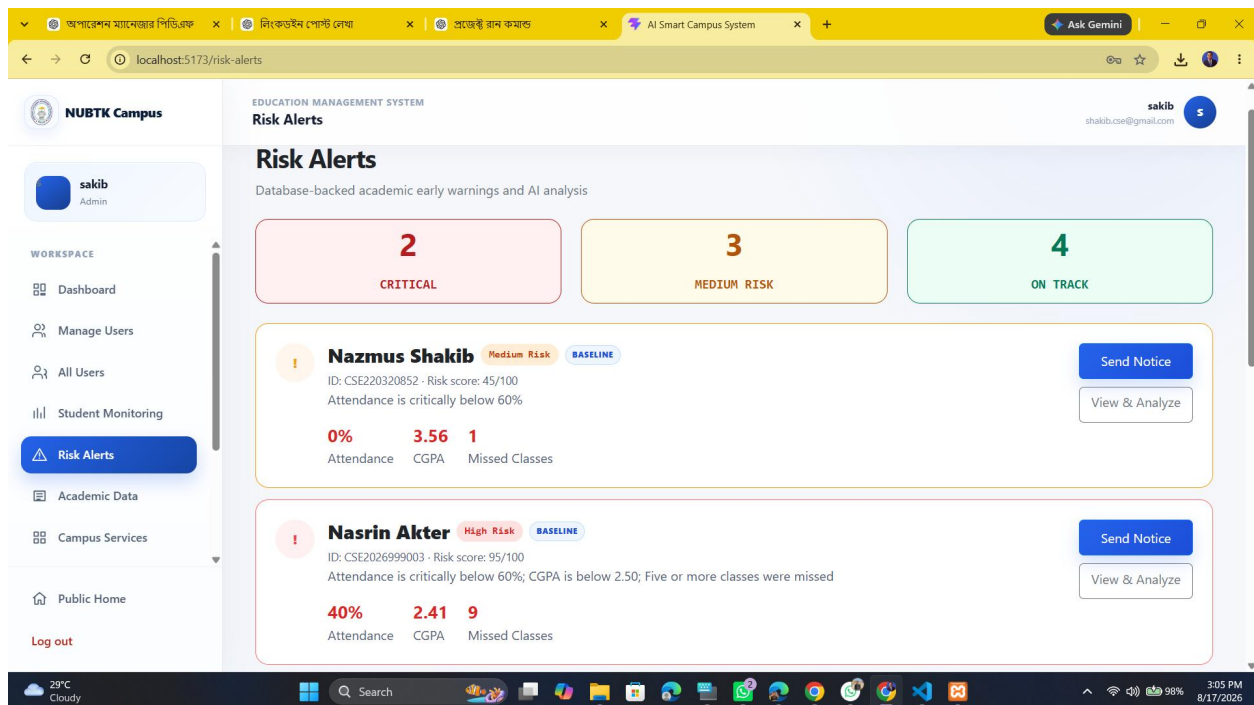


Figure 10. Database-backed academic risk alert interface with identifying card details minimized in the report.

15.5 GitHub Development Activity

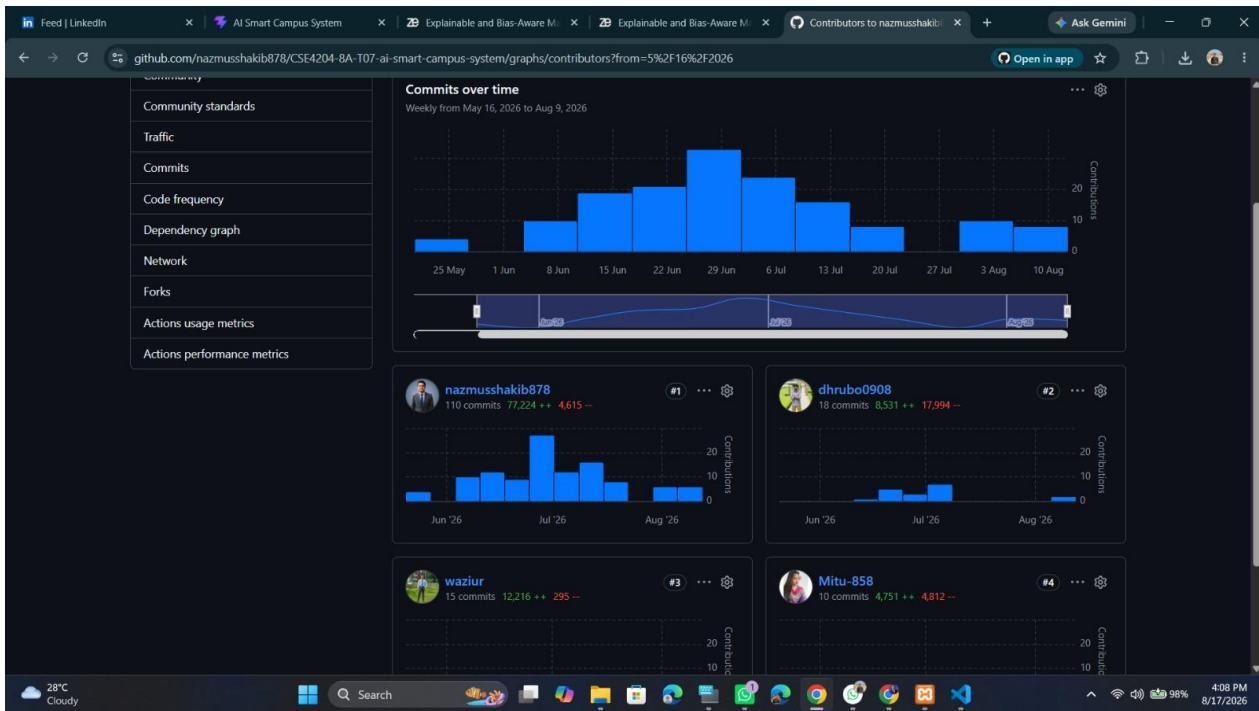
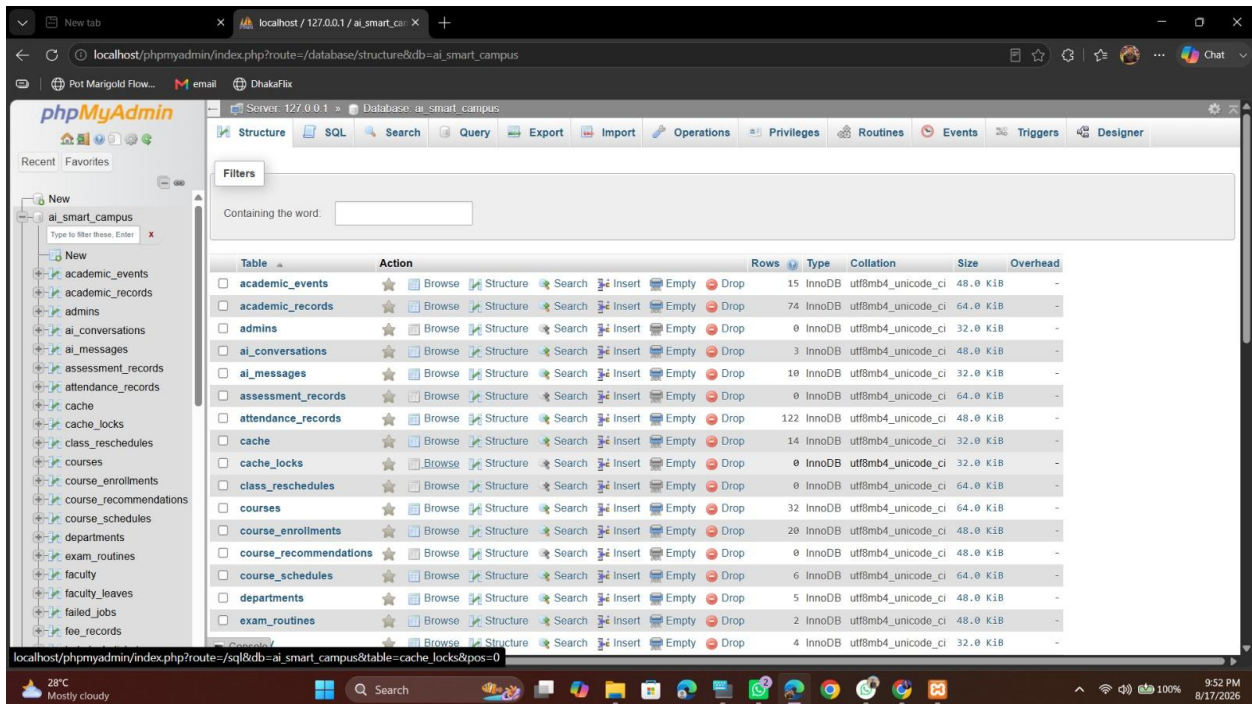


Figure 11. GitHub Insights contributors view showing recent commit activity for the project repository.

This screenshot is included exactly as supplied and provides visual evidence of continuous team contribution and recent repository activity.

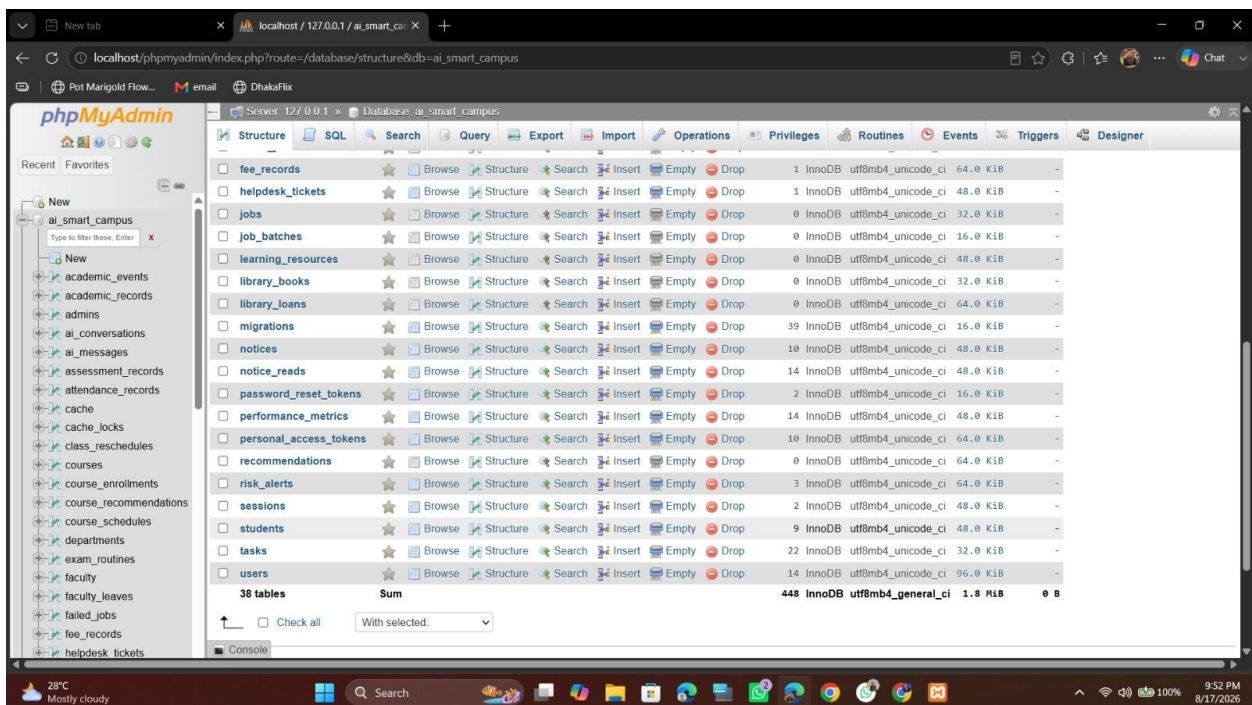
15.6 Database / Backend Evidence



The screenshot shows the phpMyAdmin interface for the 'ai_smart_campus' database. The 'Structure' tab is selected, displaying a list of tables. The left sidebar shows the database's tree structure. The main table list includes columns for Table, Action, Rows, Type, Collation, Size, and Overhead.

Table	Action	Rows	Type	Collation	Size	Overhead
academic_events	Browse, Structure, Search, Insert, Empty, Drop	15	InnoDB	utf8mb4_unicode_ci	48.0 K	-
academic_records	Browse, Structure, Search, Insert, Empty, Drop	74	InnoDB	utf8mb4_unicode_ci	64.0 K	-
admins	Browse, Structure, Search, Insert, Empty, Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 K	-
ai_conversations	Browse, Structure, Search, Insert, Empty, Drop	3	InnoDB	utf8mb4_unicode_ci	48.0 K	-
ai_messages	Browse, Structure, Search, Insert, Empty, Drop	10	InnoDB	utf8mb4_unicode_ci	32.0 K	-
assessment_records	Browse, Structure, Search, Insert, Empty, Drop	0	InnoDB	utf8mb4_unicode_ci	64.0 K	-
attendance_records	Browse, Structure, Search, Insert, Empty, Drop	122	InnoDB	utf8mb4_unicode_ci	48.0 K	-
cache	Browse, Structure, Search, Insert, Empty, Drop	14	InnoDB	utf8mb4_unicode_ci	32.0 K	-
cache_locks	Browse, Structure, Search, Insert, Empty, Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 K	-
class_reschedules	Browse, Structure, Search, Insert, Empty, Drop	0	InnoDB	utf8mb4_unicode_ci	64.0 K	-
courses	Browse, Structure, Search, Insert, Empty, Drop	32	InnoDB	utf8mb4_unicode_ci	64.0 K	-
course_enrollments	Browse, Structure, Search, Insert, Empty, Drop	20	InnoDB	utf8mb4_unicode_ci	48.0 K	-
course_recommendations	Browse, Structure, Search, Insert, Empty, Drop	0	InnoDB	utf8mb4_unicode_ci	48.0 K	-
course_schedules	Browse, Structure, Search, Insert, Empty, Drop	6	InnoDB	utf8mb4_unicode_ci	64.0 K	-
departments	Browse, Structure, Search, Insert, Empty, Drop	5	InnoDB	utf8mb4_unicode_ci	48.0 K	-
exam_routines	Browse, Structure, Search, Insert, Empty, Drop	2	InnoDB	utf8mb4_unicode_ci	48.0 K	-
fee_records	Browse, Structure, Search, Insert, Empty, Drop	4	InnoDB	utf8mb4_unicode_ci	32.0 K	-

Figure 12. phpMyAdmin database structure view showing the AI Smart Campus database tables (upper portion).



The screenshot shows the phpMyAdmin interface for the 'ai_smart_campus' database, displaying the lower portion of the table list. The 'Structure' tab is selected, showing a continuation of the table list. The left sidebar shows the database's tree structure. The main table list includes columns for Table, Action, Rows, Type, Collation, Size, and Overhead.

Table	Action	Rows	Type	Collation	Size	Overhead
fee_records	Browse, Structure, Search, Insert, Empty, Drop	1	InnoDB	utf8mb4_unicode_ci	64.0 K	-
helpdesk_tickets	Browse, Structure, Search, Insert, Empty, Drop	1	InnoDB	utf8mb4_unicode_ci	48.0 K	-
jobs	Browse, Structure, Search, Insert, Empty, Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 K	-
job_batches	Browse, Structure, Search, Insert, Empty, Drop	0	InnoDB	utf8mb4_unicode_ci	16.0 K	-
learning_resources	Browse, Structure, Search, Insert, Empty, Drop	0	InnoDB	utf8mb4_unicode_ci	48.0 K	-
library_books	Browse, Structure, Search, Insert, Empty, Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 K	-
library_loans	Browse, Structure, Search, Insert, Empty, Drop	0	InnoDB	utf8mb4_unicode_ci	64.0 K	-
migrations	Browse, Structure, Search, Insert, Empty, Drop	39	InnoDB	utf8mb4_unicode_ci	16.0 K	-
notices	Browse, Structure, Search, Insert, Empty, Drop	10	InnoDB	utf8mb4_unicode_ci	48.0 K	-
notice_reads	Browse, Structure, Search, Insert, Empty, Drop	14	InnoDB	utf8mb4_unicode_ci	48.0 K	-
password_reset_tokens	Browse, Structure, Search, Insert, Empty, Drop	2	InnoDB	utf8mb4_unicode_ci	16.0 K	-
performance_metrics	Browse, Structure, Search, Insert, Empty, Drop	14	InnoDB	utf8mb4_unicode_ci	48.0 K	-
personal_access_tokens	Browse, Structure, Search, Insert, Empty, Drop	10	InnoDB	utf8mb4_unicode_ci	64.0 K	-
recommendations	Browse, Structure, Search, Insert, Empty, Drop	0	InnoDB	utf8mb4_unicode_ci	64.0 K	-
risk_alerts	Browse, Structure, Search, Insert, Empty, Drop	3	InnoDB	utf8mb4_unicode_ci	64.0 K	-
sessions	Browse, Structure, Search, Insert, Empty, Drop	2	InnoDB	utf8mb4_unicode_ci	48.0 K	-
students	Browse, Structure, Search, Insert, Empty, Drop	9	InnoDB	utf8mb4_unicode_ci	48.0 K	-
tasks	Browse, Structure, Search, Insert, Empty, Drop	22	InnoDB	utf8mb4_unicode_ci	32.0 K	-
users	Browse, Structure, Search, Insert, Empty, Drop	14	InnoDB	utf8mb4_unicode_ci	96.0 K	-
38 tables	Sum	448	InnoDB	utf8mb4_general_ci	1.8 M	0 B

Figure 13. phpMyAdmin database structure view showing additional AI Smart Campus database tables (lower portion).

15.7 Mobile / Responsive Authentication

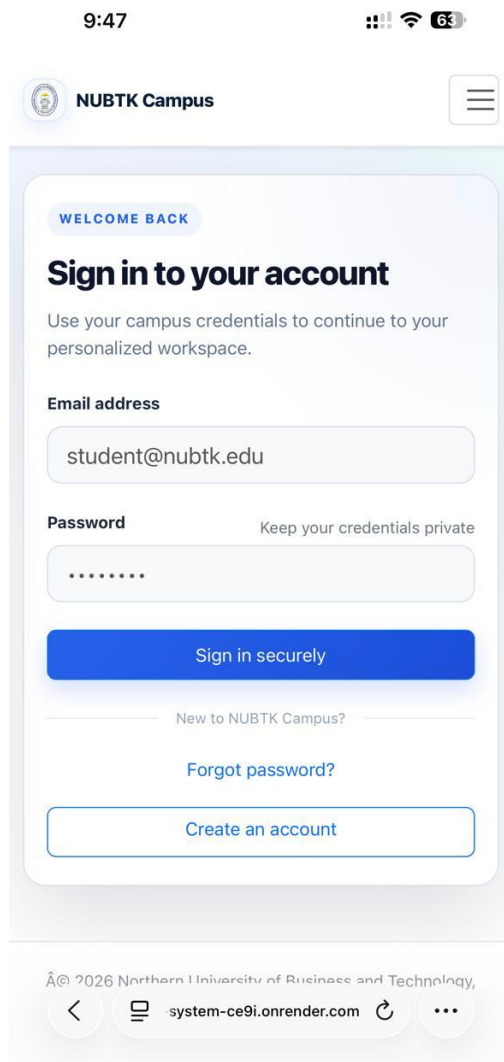


Figure 14. Mobile sign-in interface on the live Render deployment.

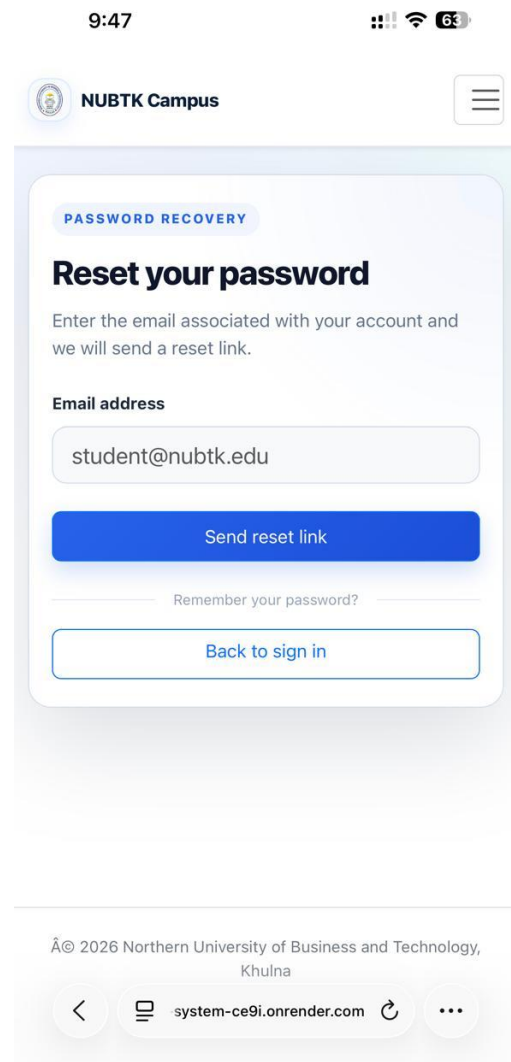


Figure 15. Mobile password recovery interface on the live Render deployment.

15.8 Mobile / Responsive Registration

9:47

NUBTK Campus

JOIN THE CAMPUS

Create your account

Set up your profile once, then access every academic tool after approval.

Full name

Nazmus Shakib

Email

you@example.com

Phone number

01712345678

Role

Student

Department

Select department

system-ce9i.onrender.com

CE66334459156

Figure 16. Mobile registration interface - profile, role, and department fields.

9:47

NUBTK Campus

Department

Select department

Student ID

CE66334459156

Password

Password

Use 8+ characters with uppercase, lowercase, a number, and a symbol.

Confirm password

Confirm password

Create my account

Already have an account? [Sign in](#)

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ai-smart-campus-system-ce9i.onrender.com

Figure 17. Mobile registration interface - department, student ID, and password fields.

15.9 Mobile Registration Completion and Public Landing

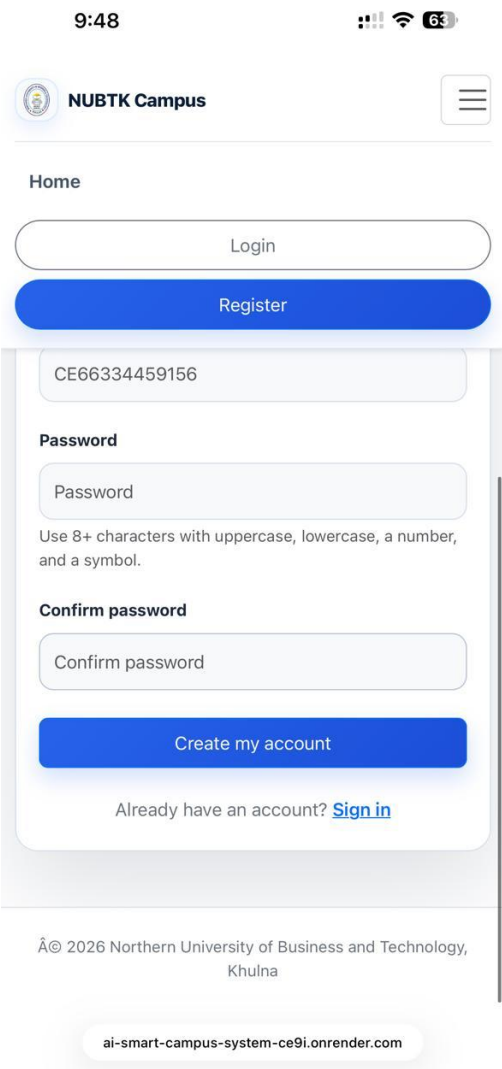


Figure 18. Mobile registration interface - password confirmation and account creation.

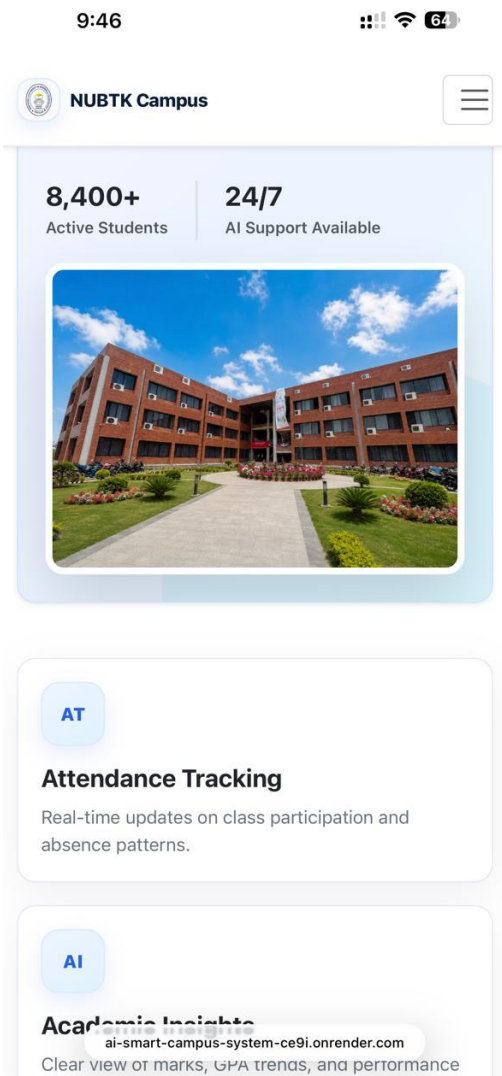


Figure 19. Mobile public landing page - active student statistics, campus image, and feature cards.

15.10 Mobile Public Landing and Call-to-Action

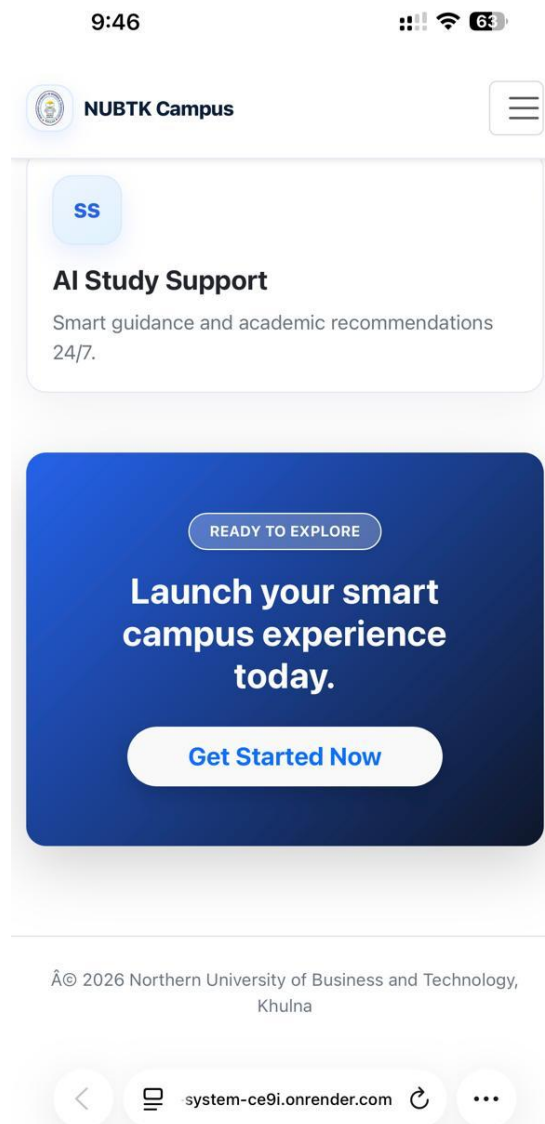


Figure 20. Mobile public landing page - AI Study Support and smart campus call-to-action.

15.11 Backend AI Integration Evidence

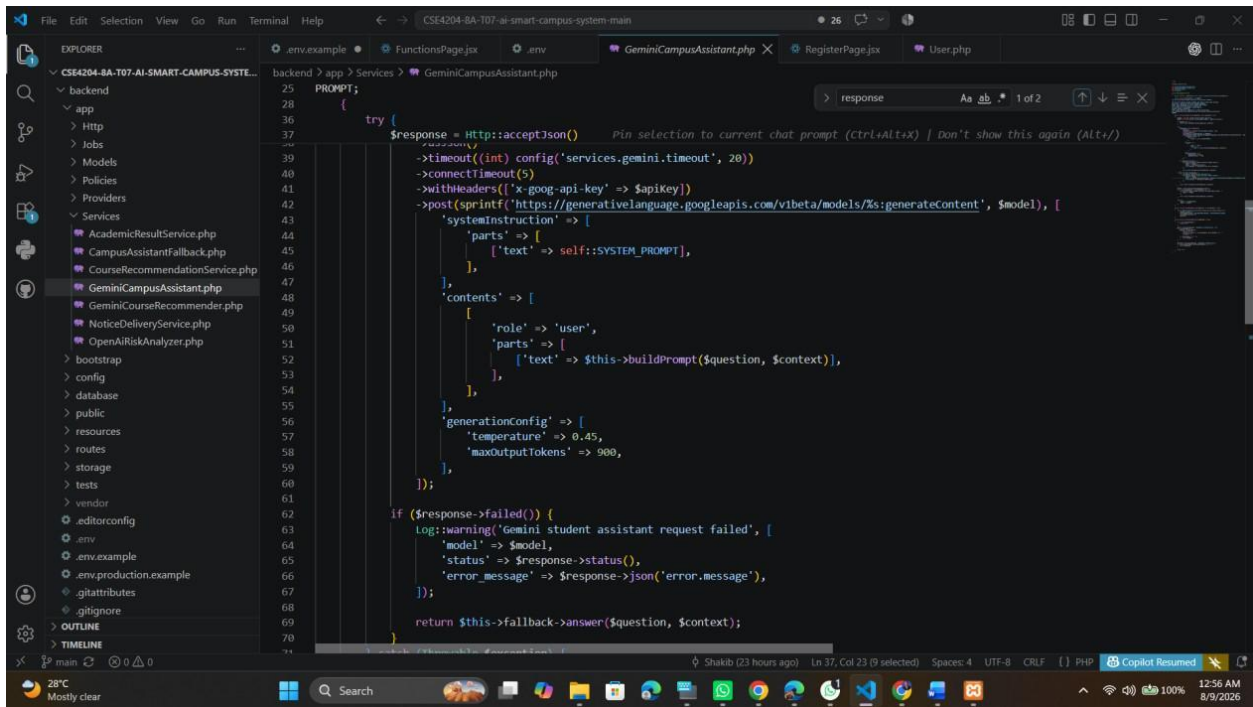


Figure 21. Backend integration evidence - Laravel GeminiCampusAssistant service constructing the server-side AI request with timeout and error handling.

16. Submission Readiness and Final Evidence Notes

The major application modules are integrated and demonstrable. The evidence in this report completes the Week 09 package, while repository-side final checks and negative-path testing should be finished before the review.

- Mobile/Responsive interface evidence is included in Figures 14-20 from the live Render deployment.
- GitHub repository/activity evidence is included in Figure 11.
- Database/backend functionality evidence is included in Figures 12, 13, and 21.
- Review every screenshot for personal student information, API keys, authorization headers, email addresses, and phone numbers before submission.
- Before Classroom submission, confirm the latest Week 09 implementation is pushed and README.md reflects the current integrated system with meaningful recent commit messages.

17. Conclusion

By Week 09, the AI Smart Campus System has progressed from separate development modules into an integrated application covering authentication, student/faculty/admin workflows, database-backed academic management, campus services, recommendations, monitoring, and AI-assisted features. The Week 09 documentation, mobile evidence, backend evidence, and major integration checks are complete. The remaining work is limited to explicit database/network negative-path evidence, repository-side final checks, and the formal testing/debugging phase.